Day 04: TCP/IP and OSI Models

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Cybersecurity Learning Log

Part 1: OSI and TCP/IP Theory

1. What is a network protocol?

A network protocol is a set of rules that devices follow to communicate and exchange data over a network.

2. What is the OSI Model?

The Open Systems Interconnection (OSI) Model is a conceptual framework that divides network communication into seven layers. It helps explain how data moves between devices and makes network troubleshooting easier.

3. Name all seven OSI layers in correct order.

From Layer 1 to Layer 7:

1. Physical
2. Data Link
3. Network
4. Transport
5. Session
6. Presentation
7. Application

4. What is the purpose of the Application Layer?

The Application Layer provides network services used by applications, such as web browsing, email, and file transfer. Examples of application-layer protocols include HTTP, HTTPS, DNS, and SMTP.

5. What is the purpose of the Transport Layer?

The Transport Layer supports communication between applications on different devices. TCP provides reliable, ordered delivery, while UDP provides connectionless communication with lower overhead.

6. What is the purpose of the Network Layer?

The Network Layer handles logical addressing and routing packets between networks. IP addresses and IP packets are associated with this layer.

7. What is the purpose of the Data Link Layer?

The Data Link Layer transfers frames over a local network link. It uses link-layer addressing, such as MAC addresses in Ethernet, and helps devices communicate on the same network segment.

8. What is the purpose of the Physical Layer?

The Physical Layer transmits raw bits through a physical medium, such as copper cables, fiber-optic cables, or wireless signals.

9. What is the TCP/IP Model?

The TCP/IP Model describes the network architecture used by the internet. Its commonly taught four layers are Application, Transport, Internet, and Network Access.

10. Explain the difference between OSI and TCP/IP models.

The OSI Model has seven layers and is mainly used as a reference framework for understanding and troubleshooting communication. The TCP/IP Model has four commonly taught layers and reflects the protocol architecture used in real-world internet communication. The OSI Session and Presentation layers are generally included within the TCP/IP Application layer, while its Network Access layer covers functions associated with the OSI Physical and Data Link layers.

11. What is encapsulation?

Encapsulation occurs when each networking layer adds its own control information to data as it moves down the protocol stack before transmission. For example, a transport layer adds a TCP header, and the network layer adds an IP header.

12. What is decapsulation?

Decapsulation is the reverse process. The receiving device processes the incoming data through its networking layers and removes or interprets the relevant headers to deliver the original data to the correct application.

13. Which layer uses IP addresses?

The OSI Network Layer, or Layer 3, uses IP addresses for logical addressing and routing.

14. Which layer uses MAC addresses?

The OSI Data Link Layer, or Layer 2, uses MAC addresses for communication on Ethernet networks.

15. Which layer uses TCP and UDP?

The Transport Layer, or Layer 4, uses TCP and UDP.

16. Which layer includes HTTP and HTTPS?

HTTP and HTTPS are Application Layer protocols, or Layer 7 in the OSI model.

17. Why should cybersecurity professionals understand networking layers?

Understanding networking layers helps cybersecurity professionals investigate suspicious traffic, interpret packet captures, identify exposed services, troubleshoot connectivity, and understand where security controls operate. It also helps them recognize attacks involving protocols, IP addresses, ports, and application traffic.

18. Explain how network data travels from a browser to a server.

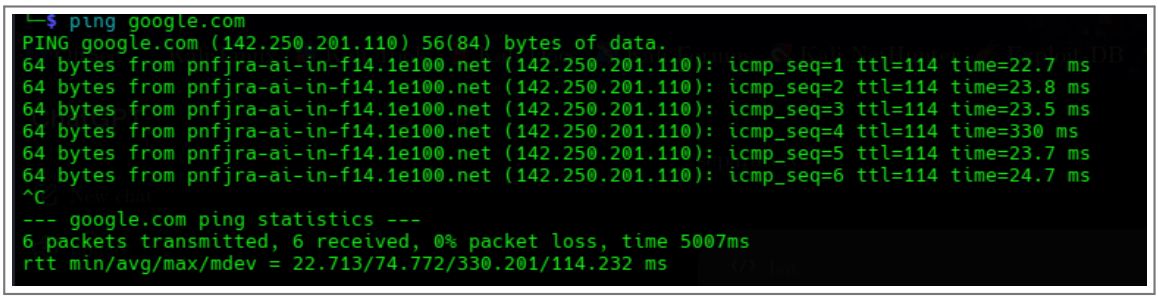
When a user enters a website address, the browser may use DNS to resolve the domain name to an IP address. The device then establishes the necessary transport connection, commonly using TCP for HTTPS over TCP, or uses QUIC over UDP for HTTP/3. The data is encapsulated into transport and IP packets and then into link-layer frames for transmission. Routers forward packets toward the server. The server processes the request and sends a response back through the network, where the receiving device processes the data and delivers it to the browser.

Part 2: Practical Results

Item	Result	Source
Resolved IP address for google.com	192.178.193.138, 192.178.193.102, 192.178.193.100, 192.178.193.113, 192.178.193.139, 192.178.193.101	Figure 3 (nslookup)
IP address used by ping	142.250.201.110	Figure 1 (ping)
DNS server used	192.168.100.1 (port 53)	Figure 3 (nslookup)
Ping result	6 packets sent, 6 received, 0% packet loss, average 74.772 ms, TTL 114	Figure 1 (ping)
Number of hops to google.com	Hops 1 to 8 replied (last reply from 10.253.4.6). Hops 9 to 14 showed no reply before the trace was stopped.	Figure 2 (tracepath)

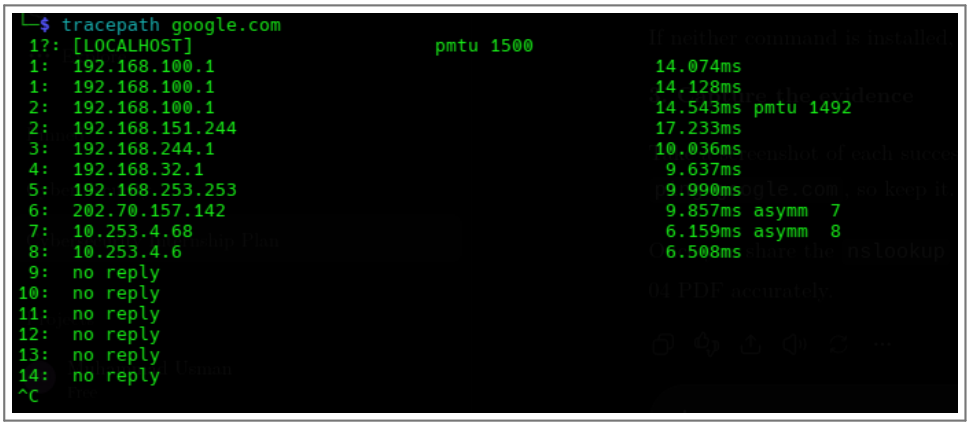
Part 3: Screenshots

Figure 1: Connectivity Test (ping google.com)



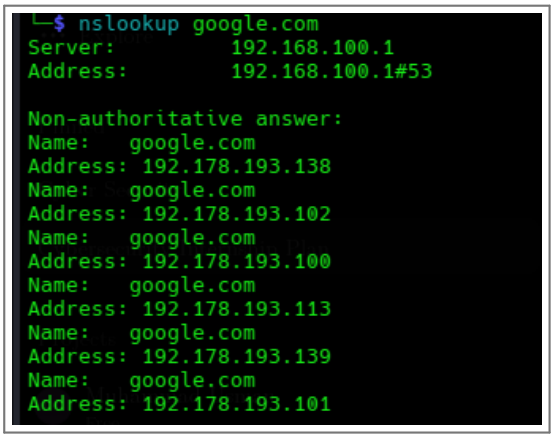
ping google.com resolved the name to **142.250.201.110**. All 6 packets were received (0% packet loss) with TTL 114. Reply times were around 23 ms, with one slower reply of 330 ms, giving an average of 74.772 ms.

Figure 2: Route to google.com (tracepath)



tracepath google.com was used on Kali Linux as the equivalent of tracert. The first hop is **192.168.100.1**, the local gateway, followed by private and ISP addresses up to hop 8 (10.253.4.6). Hops 9 to 14 show *no reply*, which is common when routers do not answer these probes.

Figure 3: DNS Lookup (nslookup google.com)



nslookup google.com was answered by the DNS server **192.168.100.1** on port 53. The answer is non-authoritative and returns six IP addresses for google.com. DNS can return different addresses depending on the network and time.

Conclusion

The OSI and TCP/IP models provide a structured way to understand how data moves across networks. Learning how protocols, addresses, ports, and network layers work together builds a useful foundation for cybersecurity monitoring, traffic analysis, and troubleshooting.